# Experiment 11: Tic Tac Toe Game

## Aim

To develop a Python program that implements the Tic Tac Toe game, allowing two players to play alternately on a 3×3 board until one player wins or the game ends in a draw.

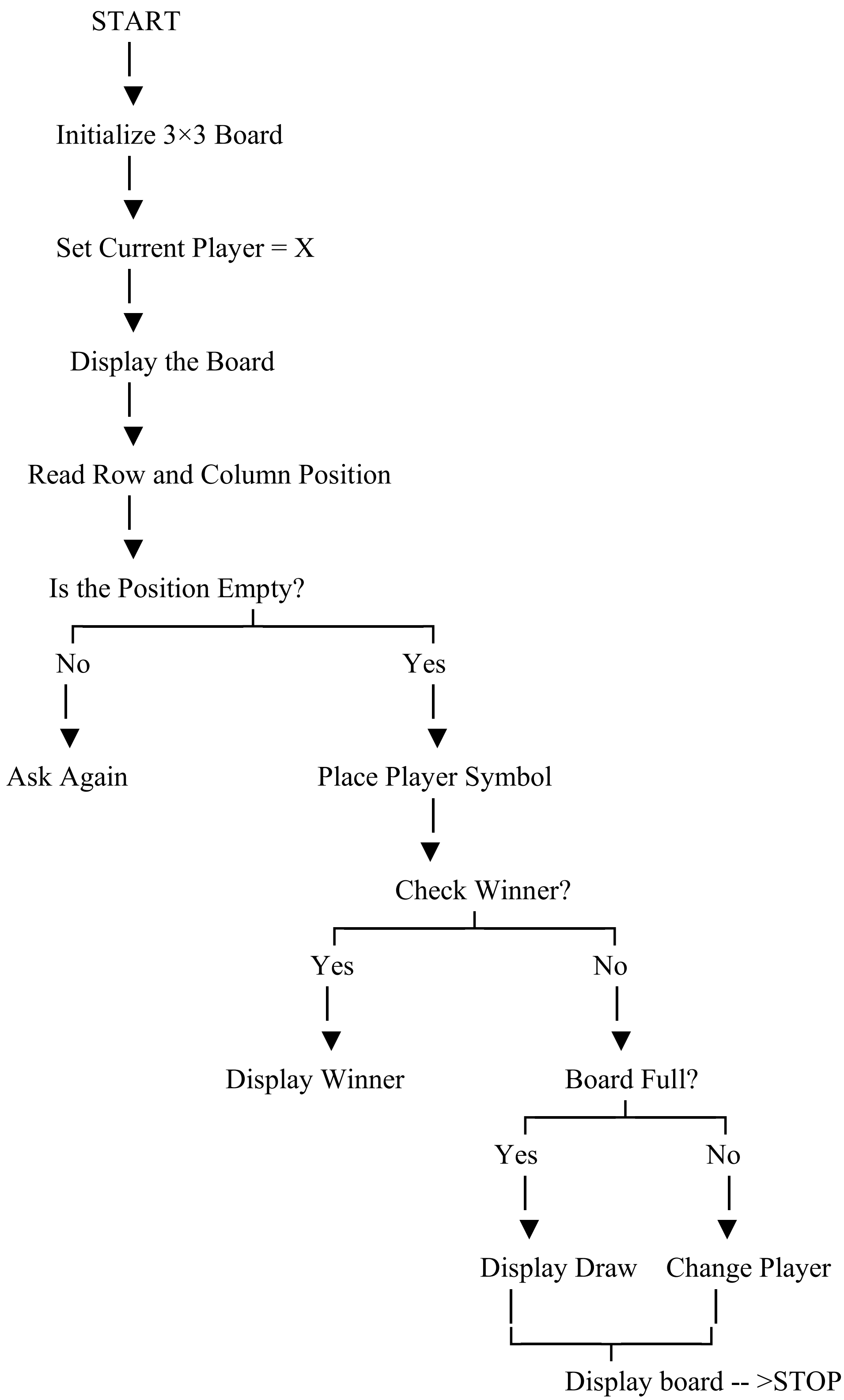
## Objective

- To understand the implementation of game-playing algorithms.
- To learn board representation using lists.
- To implement player turns and winner checking.
- To identify winning and draw conditions.

## Algorithm

1. Start the program.
2. Create a 3×3 Tic Tac Toe board.
3. Set the first player as **X**.
4. Display the board.
5. Accept the row and column from the current player.
6. If the selected position is empty, place the player's symbol.
7. Check whether the player has won.
8. If a player wins, display the winner and stop.
9. If all cells are filled without a winner, declare a draw.
10. Otherwise, switch the player and repeat from Step 4.
11. Stop.

# Flowchart

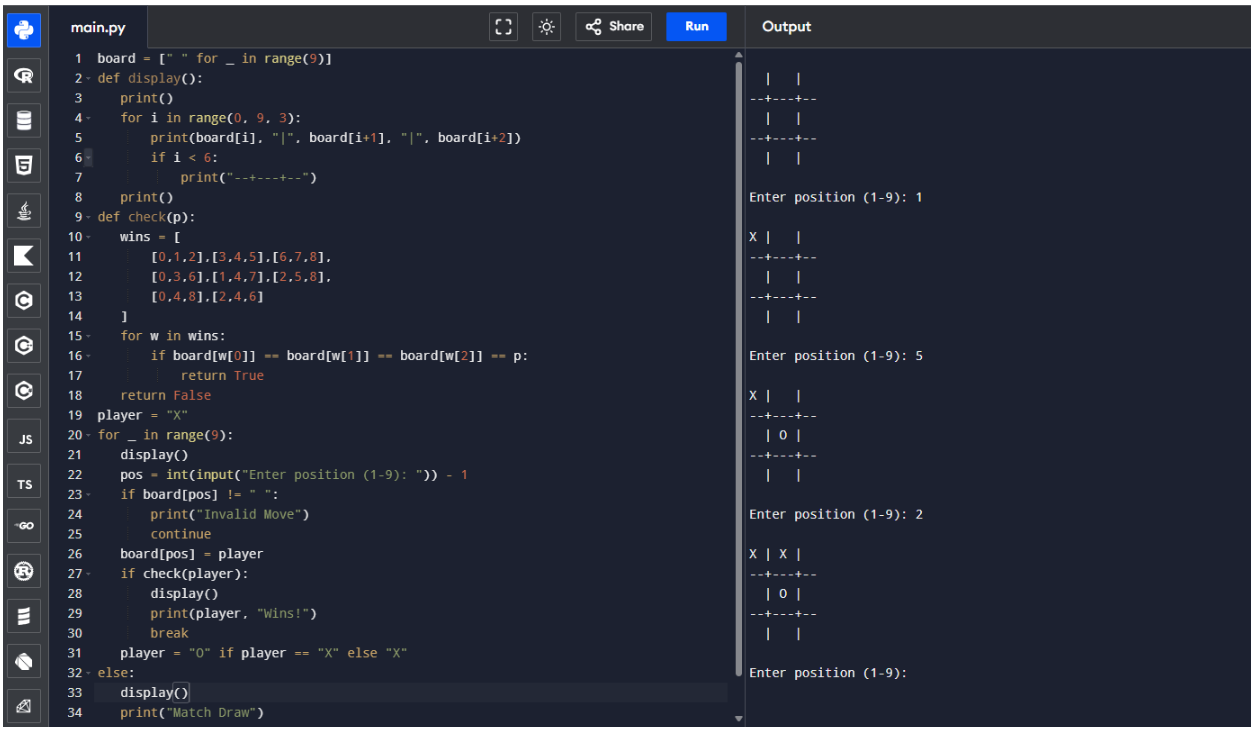

START
Initialize 3×3 Board
Set Current Player = X
Display the Board
Read Row and Column Position
Is the Position Empty?
No
Yes
Ask Again
Place Player Symbol
Check Winner?
Yes
No
Display Winner
Board Full?
Yes
No
Display Draw
Change Player
Display board -- >STOP

# Python Program

```
board = [" " for _ in range(9)]

def display():
    print()
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
        if i < 6:
            print("--+---+--")
    print()

def check(p):
    wins = [
        [0,1,2],[3,4,5],[6,7,8],
        [0,3,6],[1,4,7],[2,5,8],
        [0,4,8],[2,4,6]
    ]
    for w in wins:
        if board[w[0]] == board[w[1]] == board[w[2]] == p:
            return True
    return False

player = "X"

for _ in range(9):
    display()
    pos = int(input("Enter position (1-9): ")) - 1

    if board[pos] != " ":
        print("Invalid Move")
        continue

    board[pos] = player

    if check(player):
        display()
        print(player, "Wins!")
        break
```

```
    player = "O" if player == "X" else "X"
else:
    display()
    print("Match Draw")
```

## Sample Output




## Result

Thus, the Tic Tac Toe game was successfully implemented in Python, and the program correctly identified the winner or a draw.

## Conclusion

The Tic Tac Toe game demonstrated the fundamentals of game development, including board representation, player turns, move validation, and winner detection. It also introduced the basic concepts used in AI-based game-playing systems.